TRAN NAM ANH (LINE)

AI / MACHINE LEARNING ENGINEER | DATA ENGINEERING

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AI/ML Engineer with experience spanning machine learning research, data engineering, and LLM-based application development. Skilled in building data pipelines, developing evaluation frameworks, and benchmarking machine learning models on large-scale clinical datasets. Research experience includes Vietnamese language modeling and robust model evaluation, with an emphasis on data quality, system reliability, and practical AI engineering.

TECHNOLOGICAL SKILLS

- **Programming:** Python, SQL, TypeScript, JavaScript, R
- **Machine Learning & NLP:** PyTorch, XGBoost, LightGBM, model evaluation, model benchmarking, multi-agent systems
- **Data Engineering:** Data validation, data normalization, asynchronous pipelines, PostgreSQL, MongoDB, MySQL, vector databases
- **Backend & Infrastructure:** REST APIs, Docker, Git/GitHub, CI/CD, automated testing
- **LLM Systems:** OpenAI, Gemini, Groq, Cerebras, LLM orchestration

WORK & RESEARCH EXPERIENCE

01 | Twohearts Technologies | Data Engineer | 2026 – Present

- Built data normalization and validation workflows for AI-powered ordering systems, improving data consistency across customer, menu, and order-processing pipelines.
- Developed an internal LLM/ML evaluation harness to benchmark models, inspect execution traces, and validate Vietnamese–English tool-use workflows.
- Engineered automated end-to-end testing for AI ordering flows, combining Gherkin scenarios and property-based tests to strengthen reliability before production integration.

02 | Hospital for Tropical Diseases HCMC Collaboration | First Author / Lead AI Researcher | 2026 – Present

- Built a deterministic clinical data pipeline to extract, harmonize measurement units, and validate **3.71M+ laboratory records** across approximately **40,900 patient encounters**, under strict data privacy constraints.
- Benchmarked XGBoost and LightGBM against permutation-invariant neural architectures, including Deep Sets and Set Transformer, using **5×3 nested patient-grouped validation**.
- Achieved **0.798 AUPRC** and **0.952 AUROC** in predictive modeling experiments.
- Designed count-only, availability-only, and value-shuffled negative controls to investigate informative missingness and identify potential workflow-sensitive predictive signals.

FEATURED PROJECTS

01 | GoiBERT-M | Vietnamese Language Encoder | Preprint 2026

- Co-developed GoiBERT, a family of compact Vietnamese encoders with 26.4M and 49.4M parameters, trained through two-stage knowledge distillation on 328.6M Vietnamese tokens.
- Achieved an average score of 81.13 across eight Vietnamese NLP benchmarks, outperforming non-teacher baselines including BamiBERT (80.65) and PhoBERT-v2 (80.71) with GoiBERT-large.
- Achieved 2× GPU throughput with approximately half the parameters and peak memory of BamiBERT; GoiBERT-base retained 99% of BamiBERT's average score at one-quarter of its parameter count.

02 | Panic Hub – Real-time Agentic AI Simulation Game | AI Architect | Mar 2026

- Designed the core simulation logic for a multi-agent environment with 50+ concurrent agents, combining probabilistic decision-making, LLM-driven events, and macroeconomic constraints.
- Built a multi-provider LLM routing layer across Gemini, Groq, and Cerebras, targeting low-latency inference and cost-efficient agent interactions.
- Implemented asynchronous execution and Pydantic-based output validation to keep agent interactions structured and simulation runs robust.
- Tech Stack: Python 3.11+, PyTorch, Pydantic, AsyncIO, LLM APIs (Gemini/Groq/Cerebras)

LEADERSHIP & COMMUNITY

Google Developer Group on Campus (GDGoC HCMIU) | Vice Lead Chapter | Aug 2025 - Present

- Co-directed technical strategy and community initiatives focused on AI/ML, Cloud Architecture, and Data Analytics, engaging 100+ active student members.
- Led cross-functional teams organizing technical bootcamps, workshops, and industry expert talks.

EDUCATION & ACHIEVEMENTS

International University - VNU-HCM | B.S. in Data Science | Sep 2023 - Present

- **GPA:** 3.56 / 4.0 (86 / 100) | 2x Academic Encouragement Scholarship (AY 2025-2026)
- **Academic Role:** Teaching Assistant (TA) - Principles of Database Management
- **Awards:** Winner (KFC Track) - Agentic AI Build Week by GenAI Fund (Team Twohearts) | 3rd Place - IU AI Competition
- **Certifications:** Google Advanced Data Analytics, Google AI Essentials, Google Data Analytics Professional Certificates